AIMI: Agentic In-Hand Manipulation for Language-Guided Visual Exploration

Anonymous CVPR submission



Figure 1. **Language-guided visual exploration with AIMI.** The agent chooses how to inspect an object, uses new views to identify the requested face, and guides its presentation. Staged simulation views illustrate the task; recorded trials are reported separately.

Abstract

A language-specified target may be hidden from the camera during in-hand manipulation, requiring the robot to acquire visual evidence before presenting it. We introduce AIMI, an agentic framework that coordinates visual exploration with continuous hand control. The agent selects inspection directions and durations, uses new RGB observations and state feedback to bind the requested target to a motor goal, and guides its presentation. Teacher-supervised local controllers adjust motor commands during reasoning and execution. Controlled studies separately examine fresh visual evidence, inspection choices, and control during reasoning. Across six object-specific simulation systems, our framework achieves 65.8% mean success under changed target-to-skill mappings, exceeding the strongest evaluated baseline by 40.4 percentage points. We demonstrate sim-to-real transfer on four real-world objects using RGB pose feedback, achieving 87.5% success over 80 trials, compared with 66.25% for the strongest evaluated baseline.

1. Introduction

Dexterous in-hand manipulation lets robots reposition and inspect objects without releasing their grasp. A target specified by language may be hidden from the camera. To "present the face marked T," the robot must expose and identify it, then reach the requested orientation without dropping the object (Fig. 1).

Learning-based controllers reach specified poses or goal images [3, 4, 12]. Yet an instruction may name a surface whose motor goal is unknown. Visual evidence is needed to connect that surface to a presentation skill.

Language-based agents select skills and revise plans from feedback [9, 11]. Within a grasp, two questions arise: which view should the hand obtain next, and how should it control the object while the agent reasons?

Our goal is language-guided visual exploration with continuous in-hand control. The agent uses hand motion to gather images and link a requested visual target to a motor goal. This *target binding* guides presentation.

We implement this approach in AIMI (**A**gentic **I**n-hand **M**anipulation for **I**nspection). We adapt the ViserDex teacher–student foundation to Revo3 [4] and use teacher su-

pervision to connect the student to an agent-selectable goal bank. Local controllers adjust its joint commands during reasoning and execution. While a reply is pending, learned slowing and joint correction keep control responsive to measured feedback.

We evaluate six object-specific simulation systems, achieving 65.8% mean success under changed target-to-skill mappings. Controlled studies distinguish receiving fresh images from choosing inspection directions and durations, and separately examine control during reasoning. Multilingual trials test instruction coverage. We also demonstrate transfer to four physical objects using RGB pose feedback, with 87.5% success over 80 trials. Our contributions are:

- An agentic framework that selects inspection directions and durations, binds visual targets to motor goals, and guides their presentation through successive observations.
- Teacher-supervised local controllers that combine learned slowing and joint correction to maintain control during agent reasoning.
- Controlled studies of fresh observations, inspection action selection and waiting control, alongside evaluations across objects, target mappings and languages, and four-object real-hand validation.

2. Related Work

Agentic reasoning and embodied feedback. Language-based agents connect task descriptions to robot actions. SayCan grounds skill selection in robotic affordances [11], and Inner Monologue uses feedback to update plans [9]. Code as Policies generates programs over perception and control APIs [15], while VoxPoser and ReKep express goals through spatial value maps and keypoint constraints [8, 10]. Magma and VERA further connect multimodal or video models to robot actions [14, 21]. Model inference also takes time while the robot continues moving. Real-time chunking [5], online action correction [19], and delay-aware preference learning [24] address aspects of this problem. LiMA separates sparse intent generation from fast dexterous execution [6]. We study these coupled observation and control needs within an ongoing grasp, where the agent must obtain new views to identify a language-specified target.

Visual and goal-conditioned manipulation. Visual feedback connects object observations to dexterous actions. ViserDex uses Gaussian-splat training images for monocular pose estimation and teacher–student learning for control [4]. GoalDex learns policies conditioned on current and goal images, and uses a VLM to select visual goals from language [3]. Our focus is the exploration needed to select a goal when the requested surface is initially hidden or occluded. New observations guide target binding, while RGB pose feedback supports physical-hand control.

RL for dexterous in-hand manipulation. Simulation-based RL has enabled physical-hand manipulation through domain randomization [1], learning across object shapes [7], and online motor adaptation [16]. RotateIt adds vision and touch [17], and AnyRotate supports varied rotation axes and hand orientations [22]. SimToolReal trains a shared goal-reaching policy on procedural objects for transfer to tool use [12]. Residual learning also supports dexterous bimanual transfer in ManipTrans [13], while latency-aware RL improves generalist policies under inference delays [23]. Our controller study examines teacher-supervised slowing and joint correction around a fixed motor policy while an agent request is pending.

3. Method

AIMI uses in-hand motion to acquire visual evidence and turn it into a presentation goal. An agent decides how to inspect the object, while a recurrent motor policy and local controllers maintain continuous control. Training proceeds from a ViserDex-based teacher–student foundation adapted to Revo3, to goal-bank adaptation, and then local-controller learning (Fig. 2). We describe the cube implementation below. Other objects use their own students and maintenance controllers; their interfaces are given in the supplement.

3.1. Problem formulation

Given an object held in the hand and an instruction ℓ, the robot must find the requested surface target and present it without dropping the object. The target may initially face away from the camera or be occluded by the fingers. Hand motion therefore serves two purposes: exposing new views and reaching the presentation goal.

At step t, the agent receives an RGB image I_t, the instruction, executed-action history and measured-state summaries. It selects a skill from a fixed goal bank, represented by g_t. The motor policy uses proprioception and history x_t, measured object pose $\hat{p}_t$ and recurrent state h_t to propose joint targets u_t. Local controllers convert these proposals into commands a_t. We write $m_t = 1$ while an agent request is pending. An inspection skill does not itself declare that the target has been found.

3.2. Learning goal-conditioned motor skills

Teacher–student foundation. We build on the ViserDex teacher–student approach adapted to the Revo3 hand [4]. The pretrained teacher uses privileged simulation information and geometric goals to provide action supervision. Starting from the corresponding student, we retain compatible encoders, recurrent and actor weights, and normalization statistics. The teacher remains frozen during the following adaptation.

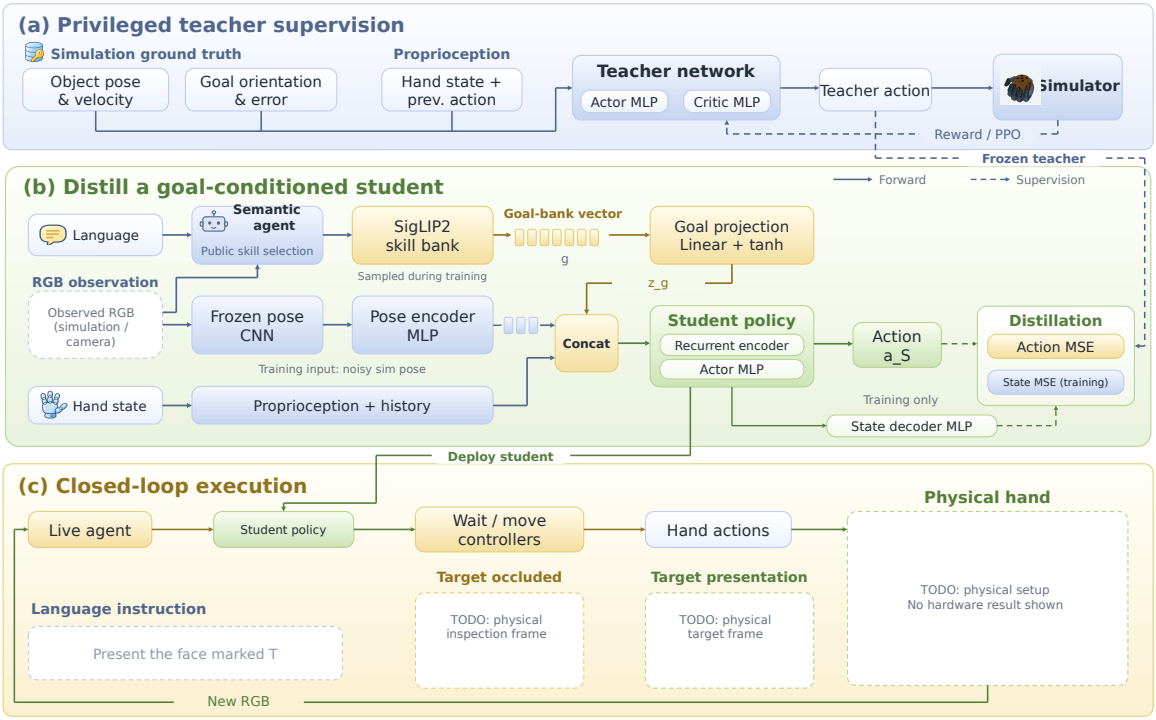


Figure 2. **AIMI training and deployment.** (a) Privileged teacher training follows the ViserDex-based Revo3 foundation. (b) The frozen teacher supervises student adaptation to goal-bank vectors; the state decoder is training-only. (c) An agent selects inspection or presentation skills from new observations while local controllers adjust joint commands. Dashed slots await physical-hand images.

Goal-bank adaptation. To connect discrete agent decisions to motor skills, we use six normalized vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [20]. Each represents a color skill and its reference object orientation. A learned projection replaces the student's geometric goal inputs:

$$z_t = \tanh(W_g g_t + b_g), \tag{1}$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \tag{2}$$

The adapted student receives neither the target quaternion nor the relative-goal quaternion. The goal bank is fixed; online agent outputs select its entries rather than supplying hidden model features.

We fit the projection and recurrent policy to frozen-teacher actions using a DAgger-style procedure [18]. Projection warmup uses action MSE. Recurrent adaptation adds a training-only decoder that reconstructs normalized privileged state with the goal quaternion removed. Its reconstruction MSE has weight 0.2. The teacher and decoder are removed from deployment. Network dimensions and fitting details are provided in the supplement.

3.3. Agentic visual exploration

The agent receives the instruction, current image, executed-action history and measured-state summaries, and returns an action: inspect, reorient, verify, stop, or abstain. State summaries include estimated skill-relative errors and current binding/request status, but no true pose or correctness label. We validate replies before applying them. Inspection selects a skill to obtain another view; target binding selects the motor goal for presentation (Fig. 3).

The agent chooses inspection and reorientation durations of 10–90 control steps. Later images reveal the effect of these actions and support the next decision. Action duration is distinct from model latency. Only one request can be pending, and the current skill remains active until a valid reply changes it. Invalid replies never select an oracle target. This loop lets the hand gather evidence before committing to a presentation goal, while control continues between decisions.

3.4. Continuous control during and after reasoning

The agent chooses goals at irregular intervals, but the hand needs commands at every control step. We therefore place local controllers around the adapted student. A sensor encoder and a goal encoder feed a fusion layer that predicts command corrections (Fig. 4). Sensor features include proprioception, measured pose, the student's recurrent state and proposal, pose-fitting error, and binding and request status.

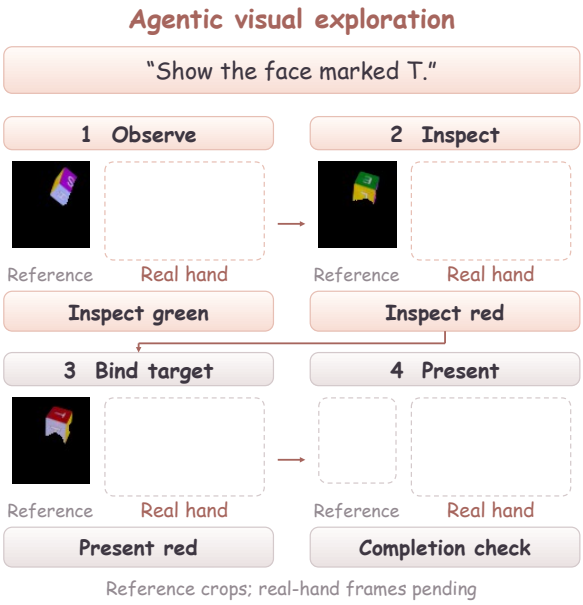


Figure 3. **Agentic visual exploration.** New observations guide inspection, target binding and presentation. Reference crops illustrate the sequence in simulation; physical-hand image slots remain open.

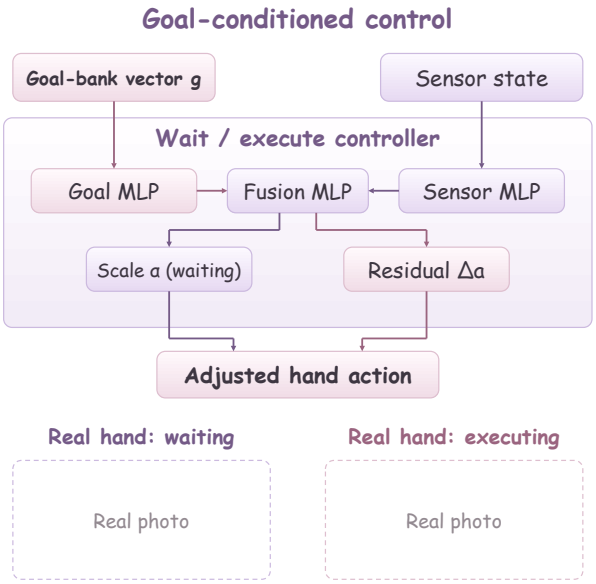


Figure 4. **Continuous local control.** Sensor features and the active goal-bank vector condition joint correction. The waiting controller also adjusts command speed. Empty slots are reserved for physical-hand images.

Waiting and execution commands. The waiting actor also receives the pose and command recorded when the request began, called the *request anchor*, and elapsed waiting time. It predicts a speed logit r_t and joint correction values v_t. With normalized current joints $\bar{q}_t$ and clipped student targets $\bar{u}_t$, the command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{3}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{4}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{5}$$

When no request is pending, a separate execution actor supplies v_t at full proposal speed. Zero waiting-actor outputs recover fixed slowing with coefficient 0.15. Joint correction can also change command direction, which scalar slowing alone cannot do. The underlying joint mapping and smoothing remain unchanged.

Teacher-supervised correction. The frozen teacher supplies holding actions for the request anchor during waiting and actions toward the active goal during execution. We train each actor against these labels with a penalty on its correction $\delta_\phi = 0.35\tanh(v_t)$:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{6}$$

Here s collects sensor features, g is the active goal and a^* is the teacher action. Before target binding, g represents a startup or inspection skill. Deployed actors use measured inputs and the active goal; they cannot access true pose, teacher actions or a pending reply. The execution actor also receives the goal, whose separate benefit is not isolated in our experiments. Controller fitting holds the student fixed, and waiting-actor updates leave the execution actor and stopping rule unchanged.

3.5. RGB perception and closed-loop deployment

For cube control, a frozen CNN takes a 120×120 RGB crop with hand occlusion and predicts image coordinates and depth for the object center and eight keypoints. Camera calibration and rigid fitting recover position, XYZW orientation and a fitting error. Following ViserDex, Gaussian-splat rendering supplies training images for this estimator [4]. The deployed model estimates pose from camera RGB.

At each control step, we estimate pose, run the recurrent student and local controller, and send joint targets. New agent observations reflect motion during inference and execution. All four physical-object systems use this RGB-to-pose-to-control interface. In simulation, cube uses RGB pose, while other objects use noisy simulator-derived pose. True pose supplies training supervision and independent scoring. Stopping and scoring are defined in Sec. 4.1.

4. Experiments

We organize the evidence around visual exploration (Q3), control during reasoning (Q2), and complete-system performance (Q1). Q3 studies visual feedback on the cube and inspection choices on the cube and pig. Q2 uses the cube. Q4 evaluates perception and four physical-object systems.

4.1. Setup and reporting

Objects and systems. Simulation uses Revo3 assets in Isaac Lab 3.0/Newton. Q1 covers a cube, die, duck, hedgehog, pig and milk carton. Instructions request one of six surface letters or a die dot count, each with a fixed face-up reference orientation. Rule maps Qwen3-VL-4B-Instruct outputs to discrete skills; Fine-tuned adapts the same backbone to predict continuous goal vectors [2]. Cube systems share the frozen RGB pose estimator, goal bank, student and stopping rule. Other objects use separate students and noisy simulator pose; AIMI adds calibrated completion correction. The VLM baselines omit the extra control actors and follow fixed inspections when the target is unreadable. Q1 compares these complete systems; Q2 and Q3 hold non-intervened components fixed within each study.

Timing and completion. Cube trials allow up to six agent calls over 120 seconds of simulated control at 30 Hz; other objects use 20 Hz. Replies cannot take effect before inference completes, and physics continues during model latency. Simulation runs slower than real time, so these rates are not hardware-speed measurements. The cube stopping check runs at 3 Hz and requires target binding, no pending request or inspection, and measured orientation error at most 0.16 radians. It may terminate without an explicit agent stop action. Independent scoring requires the correct binding, true full-quaternion error at most 0.16 radians at stop, and no drop. There is no extra holding period. Non-cube methods use the same scoring tolerance with object-specific stopping rules. Exact timing, drop boundaries and scoring equations are in the supplement; Q4 gives the physical-hand criterion.

Trials and reporting. Unless specified otherwise, each simulated condition has 40 trials, 20 per evaluation seed, matched by initial physical state and scored target. Q2 uses three seeds with 40 trials each and reports all 120 outcomes per method. Q1 and Q3 use common service-eligible trials: a service error or call exceeding 120 seconds excludes the matched trial from every method. All raw outcomes are retained without replacement. Q1's complete-system trials differ from Q2's controller cohort, so their absolute rates need not match. The supplement provides study-specific seeds, pairing details and statistical tests.

4.2. Q3: Visual Exploration and Feedback

Three interventions examine how the agent acquires and uses evidence (Table 1): continued feedback, fresh images, and inspection direction/duration. Each comparison uses its own trial set; the inspection intervention is evaluated separately on cube and pig.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98
(c) Inspection policy: cube				
Fixed sweep	31/40	77.5	1	126
Adaptive inspection	38/40	95.0	0	117
(c) Inspection policy: pig follow-up				
Fixed sweep	29/40	72.5	2	125
Adaptive inspection	37/40	92.5	2	106

Table 1. **Q3: visual evidence and inspection choices.** Each comparison schedules 40 pairs. (a,b) use cube; (a) excludes one service-affected pair and reuses Q2's initial feedback subset. (b) changes only agent RGB. (c) changes inspection direction/duration within each object system; the objects are not pooled. (b,c) have no exclusions. Drops/calls include all trials.

(a) Does continued feedback help find the target? An initial plan selects up to six bounded actions from one image and measured feedback; continued feedback allows up to six calls with new images and history. Local controllers, stopping and time limits remain fixed. Success rises from 6/39 to 33/39 eligible trials, with all 27 discordant pairs favoring feedback (exact $p = 1.49 \times 10^{-8}$). Feedback resolves 22 of 28 initial-plan abstentions; 32 trials first bind after inspection (Fig. 5). The next comparison holds the call budget fixed.

(b) Does the next image provide useful evidence? Both conditions use the same six-call cap, request rule, live numeric feedback, history and local control. Only RGB changes: the current image versus the exact first frame. Success is 32/40 versus 9/40, with all 23 discordant pairs favoring fresh images (exact $p = 2.38 \times 10^{-7}$). Correct terminal bindings number 34 versus nine. Both first bind correctly on call one in the same nine trials; fresh images enable 25 additional correct bindings, 23 leading to success, versus none with repeated frames. Drops differ by one; initial visibility is not independently annotated.

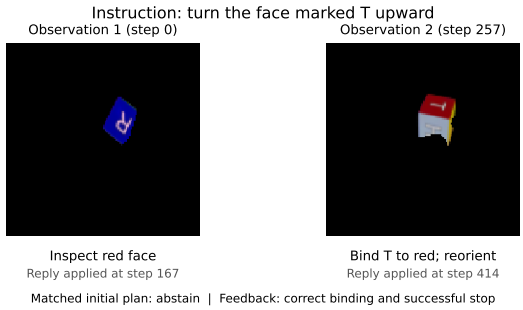


Figure 5. **New evidence enables target binding.** Unretouched model-input frames from a simulated trial. The initial plan abstains; continued feedback exposes T, after which AIMI binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from the same state and image. This selected example illustrates inspection before binding.

(c) Does choosing inspection actions help? Within each object, 40 new pairs compare agent-selected directions and durations with a prescribed color sweep of 3-second inspections. Both arms share the model, controllers, fresh-image input, history format and stopping; history records executed actions. The agent still decides whether to inspect, bind or abstain.

On cube, adaptive inspection achieves 38/40 successes (95.0%) versus 31/40 (77.5%). Nine pairs favor adaptive and two favor fixed inspection (exact $p = 0.0654$); this trend remains inconclusive. A separate pig follow-up, frozen before its formal trials, gives 37/40 (92.5%) versus 29/40 (72.5%). Eight pairs favor adaptive and none favor fixed inspection: a 20-point gain (exact $p = 0.0078$). These per-object tests are unadjusted.

Pig has 37 versus 29 correct terminal bindings, independently reconstructed from applied actions, and two drops per arm. Adaptive inspection uses 15.2% fewer calls and 26.1% fewer tokens. This supports inspection choice under the prescribed pig comparison; it does not establish faster completion. The supplement reports seed splits, binding trajectories, costs and differences between the two object systems.

4.3. Q2: Control During Agent Reasoning

Figure 6 compares four ways to control the hand while the agent is reasoning: continue the student's command, slow it by a fixed coefficient of 0.15, learn the slowing coefficient, or learn slowing together with a bounded joint correction. All four use the same execution controller when no request is pending. Each method has 120 trials: 40 under each of three evaluation seeds, with the trained weights fixed. These seeds vary evaluation tasks, not training initialization.

Learned slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning speed alone gives 82/120 successes (68.3%), below fixed slowing. The results favor learning speed and joint correction together. Drop stages tell us when failure occurs, but do not identify its cause. Seed settings and outcome breakdowns are in the supplement.

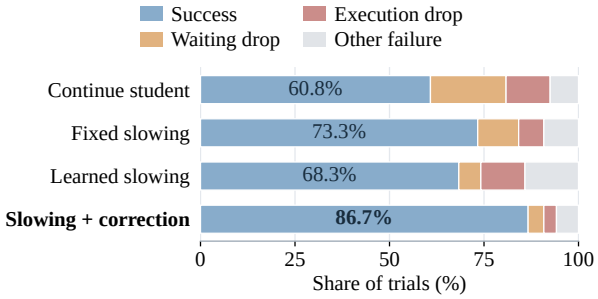


Figure 6. **Q2: outcomes during controller ablation.** Each bar contains 120 trials, 40 per evaluation seed. Labels show success rates; colored segments partition all outcomes. Drop stages record when failure occurs. Full counts are in the supplement.

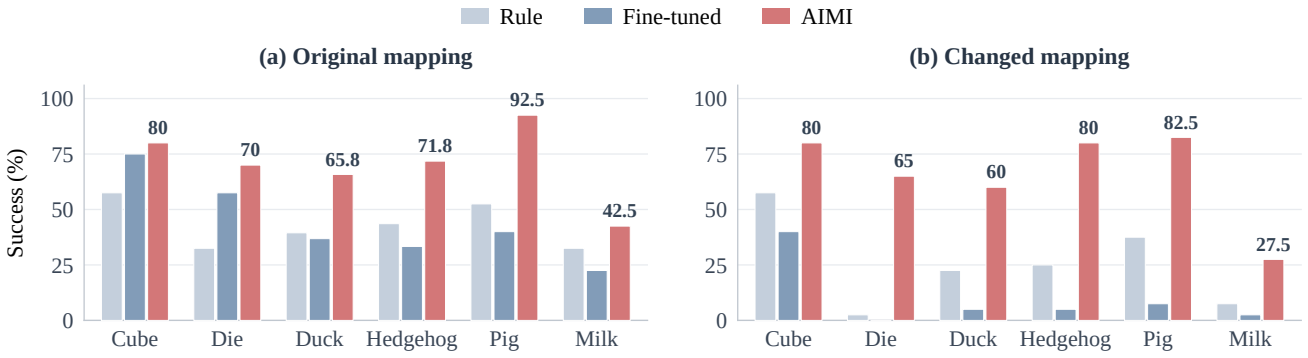


Figure 7. **Q1: task success across six simulated objects.** Bars use common service-eligible trials: 40 per cell, except 38 for original-mapping duck and 39 for original-mapping hedgehog. Labels show AIMI rates. Each object uses its own policy. The supplement retains all values and system differences. These are complete-system comparisons.

4.4. Q1: Objects, Target Mappings and Language

We compare original and changed target-to-skill mappings on six objects (Fig. 7), then vary language and wording. Separate native-image comparisons remain in the supplement.

Aggregate system performance. Across 1,440 scheduled trials, mean success is 70.4% with original mappings and 65.8% with changed mappings, exceeding the strongest baseline means (44.2% and 25.4%) by 26.2 and 40.4 points. These descriptive, post-hoc means weight each object's service-eligible rate equally.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials. AIMI leads in each condition, although milk-carton drops remain frequent: 15/40 original-mapping and 20/40 changed-mapping trials. Three service failures exclude matched trials across methods; reset checks and sensitivity analyses are in the supplement.

Visual-semantic intervention. Only the agent's RGB changes across 40 matched states with original, reassigned or hidden letters; pose-estimator input and lighting remain fixed. Reassignment leaves AIMI at 32/40 successes but reduces fine-tuning from 30/40 to 16/40 (Holm $p = 0.0130$, six comparisons); rules remain at 23/40. Hiding letters gives 0/39, 3/40 and 0/40, respectively. Nine of eleven fine-tuned wrong bindings follow the old mapping. One reassignment does not establish broad generalization.

Dice layout transfer. Across 240 die trials, rotating the visual mesh preserves collision geometry, inertia and the skill-ID prompt. Exploratory paired tests favor AIMI over both baselines under the changed layout; its own layout difference remains inconclusive. The supplement reports all seven unadjusted contrasts.

Language and wording coverage. Standard and rewritten English and Chinese instructions are tested on three objects: 60 trials per condition across three seeds, totaling 2,160 (Table 2). Weights, mappings and scoring remain fixed, without Chinese retraining. Rule also translates its task/output instructions; raw outcomes are in the supplement.

	English		Chinese	
System	Standard	Rewrite	Standard	Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
AIMI	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
AIMI	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
AIMI	**73.3**	**76.7**	**75.0**	**76.7**

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

AIMI exceeds fine-tuning in all eight duck/hedgehog conditions and rules in six after Holm correction over 36 tests. Cube comparisons remain inconclusive; fine-tuning leads for rewritten Chinese. Strict state/image pairing preserves these decisions. Language/wording contrasts remain inconclusive after a separate 36-test correction.

4.5. Q4: Pose Estimation and Sim-to-Real Transfer

Perception adaptation for transfer. We adapt the ViserDex-based perception pipeline with real-camera calibration and Gaussian-splat appearance assets. Table 3 evaluates RGB and RGB-D variants, showing improved pose recall with RGB-D in simulation. Training and camera adaptation differ between variants. This supporting perception study is separate from the RGB-pose hardware evaluation below.

Estimator	Pose recall (%) ↑		Invalid ↓
	10 mm / 10°	5 mm / 5°	
RGB	37.53	14.63	69
RGB-D	**64.73**	**47.56**	**1**

Table 3. **Q4: independent pose estimation.** Recall requires both errors below the listed thresholds. All 6,144 simulated observations remain in the denominator, including invalid predictions. Evaluation uses ideal masks and virtual depth.

Complete-system real-hand evaluation. We evaluate Rule, Fine-tuned and AIMI on a cube, die, ball and duck, with 20 trials per object and method (240 total). All use RGB-based pose estimation. A trial succeeds when the hand reaches the requested target and stops within 15° of its target orientation without dropping the object. Table 4 reports per-object success counts and rates. Overall rates are 40.0%, 66.25% and 87.5%, respectively. AIMI completes 70/80 trials, exceeding fine-tuning by 21.25 percentage points.

Object	Rule	Fine-tuned	AIMI
Cube	9/20 (45%)	15/20 (75%)	**18/20 (90%)**
Die	8/20 (40%)	14/20 (70%)	**18/20 (90%)**
Ball	9/20 (45%)	13/20 (65%)	**19/20 (95%)**
Duck	6/20 (30%)	11/20 (55%)	**15/20 (75%)**
Total	32/80 (40%)	53/80 (66.25%)	**70/80 (87.5%)**

Table 4. **Q4: real-hand task completion.** Successes/trials and rates for the four physical objects. Every object–method cell contains 20 trials. Success requires reaching the requested target and stopping within 15° without dropping the object.

5. Discussion and Limitations

Exploration and control. Q3 separates receiving new evidence from choosing how to acquire it. Fresh images support target binding on cube; the pig follow-up supports adaptive inspection under its prescribed comparison, while the cube inspection result remains inconclusive. Q2 supports joint correction during reasoning. Q1 compares complete systems differing in models, history, controllers and stopping.

Evaluation limits. Fewer calls and tokens on pig do not establish faster completion or fewer drops. Inspection comparisons cover one prescribed sweep and six targets on each of two objects. The latent interface and controller-goal conditioning lack independent ablations. Initial readability lacks independent human annotation; binding and terminal pose do not establish recognition accuracy or post-stop holding.

Deployment scope. Each object uses separate weights. Simulation uses RGB pose for cube and noisy simulator pose otherwise; hardware uses RGB pose. Hardware results do not establish unseen-object transfer or isolate perception gains. Real-camera pose accuracy remains unmeasured. Hosted-model changes limit reproducibility; service-eligible rates do not measure end-to-end availability.

6. Conclusion

AIMI connects language-guided visual exploration to continuous in-hand manipulation. Controlled studies distinguish fresh visual evidence, inspection action selection and local control during reasoning. The pig comparison supports adaptive inspection over the prescribed sweep; the cube comparison remains inconclusive. Evaluations cover six simulated objects, two languages, and four physical objects with 87.5% success over 80 hardware trials.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

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